

Portfolio-Conditioned Dynamic Reinforcement Ranking in Inverter-Rich Power Networks: Evidence from Two Benchmarks

Jorge L. Mayorga Taborda

Abstract—When synchronous generators (SGs) are replaced by grid-following converters (GFLs) unit by unit, a planner must decide which branch to reinforce in the portfolio at hand. We test the re-equilibrated (total) eigenvalue sensitivity of the planned portfolio, a known construction, as a ranking of finite branch reinforcements on two benchmarks and with two converter models, under preregistered holdout designs. On the IEEE 39-bus system its median Spearman correlation with the finite effect is 0.996 on 24 new policies, against 0.712 for the frozen-operating-point derivative and at most 0.445 for static strength indices. On the IEEE 68-bus system with governors and rotor damping it reaches 0.996 with the same GFL and 0.981 with a frozen WECC library GFL, against 0.20 and 0.43 for static indices selected on discovery conditions. There the preregistered target is a slow control pole that no reinforcement moves by more than $1.4 \times 10^{-3} \text{ s}^{-1}$; a ranking of the rightmost electromechanical mode, defined post hoc, reaches 0.998. The IEEE-39 study also found that the same replacement moves the same electromechanical mode in opposite directions in two nested stable portfolios (17/19 base-stable holdout policies, median 0.013 s^{-1}). This does not replicate on IEEE-68: with either converter, with or without governors and damping, no replacement materially stabilizes the tracked mode in any portfolio, and a fixed unit ranking has zero regret. Exact line lists differ between converter models; the ranking method carries over.

Index Terms—Grid-following converters, small-signal stability, eigenvalue sensitivity, grid strength, replacement planning, set functions.

I. INTRODUCTION

GRID-STRENGTH indices summarize how well a network supports converter-interfaced resources. Single-infeed short-circuit ratios yield context-free lists of weak buses; multi-infeed generalizations—the site-dependent SCR [1], the generalized SCR [2]–[4], the generalized operational SCR [5], small-signal strength of all-inverter systems [6] and impedance metrics [7], [8]—are defined for a given set of converter buses and are thus portfolio-conditioned static screens. Modal participation [9], [10] and impedance-based eigenvalue sensitivities [11] rank dynamic criticality.

Retirement poses a sequential question. Units are replaced one at a time, and the planner must decide which unit to replace next *given* those already replaced, and which branch

to reinforce *given* the resulting portfolio. The object is the contextual marginal

$$\Delta_i \alpha_{\perp}(S) = \alpha_{\perp}(S \cup \{i\}) - \alpha_{\perp}(S), \quad (1)$$

the change of the rightmost transverse eigenvalue’s real part when unit i is replaced in portfolio S . It has long been known that displacing SGs by converters improves some electromechanical modes and degrades others [12], that rising converter penetration changes the participation in inter-area modes and adds converter-dominated modes [13], that the small-signal effect of converters depends on the mix and dispatch in which they operate [14]–[17], and that a branch reinforcement can reduce linear stability (Braess-type behavior [18], [19]; operating-point-dependent edge weights [20]). Eigenvalue sensitivities with a re-solved operating point are classical [21]–[24], and a recent preprint chains them to line admittances [25]; topology switching changes small-signal margins [26], [27]; and natural dynamic set functions are known to violate sub- or supermodularity [28]–[30].

This paper is an empirical study in two preregistered stages: a hardening campaign on the IEEE 39-bus system and a confirmatory replication on the IEEE 68-bus system with documented governors and damping, each with two converter models. Its findings, each stated with its limits, are:

- C1) *Portfolio-conditioned reinforcement ranking on two benchmarks.* The re-equilibrated sensitivity of the planned portfolio ranks finite reinforcements better than static strength indices on both benchmarks and under both converter models, and on IEEE-39 also better than the frozen derivative and a fixed list learned from other conditions. On IEEE-68 the preregistered target is an immaterial control pole; the electromechanical (EM) version of the ranking is post hoc.
- C2) *When re-equilibration matters.* The operating-point term vanishes for controller coordinates and matters for network coordinates: on IEEE-68 the frozen derivative reaches median Spearman 0.55 against 0.996, and the two disagree in sign for 33 % of branches in the median condition.
- C3) *Contextual reversal is benchmark-specific.* On IEEE-39 the same replacement moves the same tracked EM mode in opposite directions in two nested stable portfolios in 17/19 base-stable holdout policies, at about one hundredth of s^{-1} ; an oracle fixed unit ranking errs mainly at

TABLE I
DEVICE AND BENCHMARK DATA (PER UNIT ON DEVICE RATING)

SG	two-axis, $D = 0$, constant P_m , M 4.9–8.4 s (bus 39: 100 s); AVR $K_A \in \{10.1, 40\}$, T_E 0.25–0.95 s by unit; PSS $K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,pll} = 53$, $k_{i,pll} = 1400$
GFL loops	$\tau = 0.03$ s; outer 0.2/8; inner 0.25/6; $x_f = 0.15$, $r_f = 0.01$
Q/V	$k_{p,v}/k_{i,v} = 2/20$, leak $w = 0.05$ rad/s, gain $g \in [0, 1]$
Candidates	$V_9 = \{30, \dots, 38\}$ (2^9 portfolios); core $V_4 = \{30, 33, 35, 37\}$

singularity crossings. On IEEE-68 neither finding replicates (0/16 and 0/16 policies).

- C4) *Boundaries*. Exact line lists and the best static comparator depend on the converter model; single branch actions on IEEE-39 remove and create EM incompatibilities; removing governors or damping on IEEE-68 does not create reversals.

The study is phasor-domain small-signal analysis; no electromagnetic-transient claim is made.

II. PROBLEM FORMULATION

A. Model

The network is the IEEE 39-bus system [31] (46 branches, ten SGs, bus 39 an interconnection equivalent) in phasor form with constant-power loads. The system is the index-one DAE $\dot{x} = f(x, z; p)$, $0 = g(x, z; p)$ with device states x , rectangular bus voltages z and parameters p ; at an equilibrium the reduced Jacobian is $A = f_x - f_z g_z^{-1} g_x$. Each SG is a two-axis machine [32] with $D = 0$, constant mechanical power, a first-order AVR $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$ and a power-input stabilizer. Machine, AVR and PSS data, ratings and the power flow are taken from the IEEE 39-bus dynamic case distributed with ANDES [33] (GENROU records used in two-axis form, IEEEEX1 gain K_A and time constant T_E , IEEEEST stabilizer); Table I lists the controller values. Each GFL has a synchronous-reference-frame PLL, filtered power measurements, PI outer and inner loops behind a filter reactance, and a leaky Q/V regulator $q_{\text{cmd}} = k_{p,v} g e + x_v$, $\dot{x}_v = k_{i,v} g e - w(x_v - q_{\text{ref}})$, $e = v_{\text{ref}} - |V|$, $w = 0.05$ rad/s. Converter current limits, DC links and delays are not modeled.

A policy $\theta = (g, k, t, h)$ sets the Q/V gain g of every GFL, a common scale k of the surviving SGs' AVR gains, a scale t of their AVR time constants, and a heterogeneity exponent h with $T_{E,i} \mapsto G(T_{E,i}/G)^h$, G the geometric mean. Replacing unit i installs a GFL with the SG's active and reactive dispatch and rating (matched dispatch), so the base power flow is the same for every portfolio. The flagship policy P4 = (0.03625, 1.425, 1.5, 1) makes V_4 unstable with a 0.62 Hz critical mode while all its subsets are stable.

B. Transverse stability

With $D = 0$ and no governors the model has an exact rotation symmetry and a neutral frequency mode (two structural zero eigenvalues). Stability is evaluated on the transverse

quotient that removes them; $\alpha_{\perp}$ is the largest real part of the transverse spectrum. A portfolio is STABLE if all transverse eigenvalues satisfy $\text{Re } \lambda \leq -2 \times 10^{-3}$, UNSTABLE if one has $\text{Re } \lambda \geq 2 \times 10^{-3}$, and unresolved otherwise.

C. Contextual marginals and modal levels

For a policy θ , $f(S) = \alpha_{\perp}(S; \theta)$ on $S \subseteq V_9$. A marginal (1) is *material* if $|\Delta_i f(S)| \geq \tau_{\text{mat}} = 0.01 \text{ s}^{-1}$ (about 0.2 % damping ratio at 0.7 Hz); its sign class is -1 (stabilizing), 0 or $+1$. Because a change of $\alpha_{\perp}$ can be a jump between modes, each marginal (S, i) is graded: (A) S is STABLE; (B) the critical mode of S has a match in $S \cup \{i\}$ that is the critical mode of $S \cup \{i\}$; (C) B, with both critical frequencies in 0.1–2.0 Hz; (D) for a pair of contexts, C at both, and their critical modes match each other. Matching uses the modal assurance criterion ($\text{MAC} \geq 0.8$) on the 78 rectangular bus-voltage components of the mode shape (unit norm, phase fixed at the largest entry), among eigenvalues in 0.1–2.0 Hz with $\text{Re} \geq -1 \text{ s}^{-1}$ plus the critical one, with $|\Delta f| \leq 0.15$ Hz. Level D is the “same mode” statement.

III. MARGINALS AND SECOND DIFFERENCES

For $S \subseteq V$ and distinct $i, j \notin S$ let $d_{ij}(S) = f(S \cup \{i, j\}) - f(S \cup \{i\}) - f(S \cup \{j\}) + f(S) = \Delta_i f(S \cup \{j\}) - \Delta_i f(S)$, the one-step second difference (interaction term). f is submodular if $\Delta_i f(S) \geq \Delta_i f(T)$ for $S \subseteq T$, $i \notin T$, equivalently $d_{ij}(S) \leq 0$ everywhere, and supermodular with the inequalities reversed [34]–[37]. The following algebra is classical; we use it to locate interactions, not as a new result.

Lemma 1 (chain identity). *Let $S_0 \subseteq V$, $j_1, \dots, j_m \in V \setminus S_0$ distinct, $S_r = S_{r-1} \cup \{j_r\}$, and $i \notin S_m$. Then $\Delta_i f(S_m) = \Delta_i f(S_0) + \sum_{r=1}^m d_{ij_r}(S_{r-1})$.*

Proof: $d_{ij_r}(S_{r-1}) = \Delta_i f(S_r) - \Delta_i f(S_{r-1})$; sum over r . ■

A *nested reversal* of i is a pair $S_1 \subsetneq S_2$, $i \notin S_2$, with $\Delta_i f(S_1)$, $\Delta_i f(S_2)$ in opposite sign classes; its direction is $s \rightarrow d$ (stabilizing at S_1 , destabilizing at S_2) or $d \rightarrow s$.

Proposition 1. *A nested $s \rightarrow d$ reversal is incompatible with submodularity and a nested $d \rightarrow s$ reversal with supermodularity; along the chain from S_1 to S_2 some term satisfies $d_{ij_r} \geq D/m$ if $D = \Delta_i f(S_2) - \Delta_i f(S_1) > 0$ ($\leq D/m$ if $D < 0$), $m = |S_2 \setminus S_1|$.*

Proof: Under submodularity all terms of Lemma 1 are nonpositive, so the marginal cannot increase along a growing chain; supermodularity is symmetric. The m terms sum to D , so the largest is at least D/m . ■

The bound holds for every nested pair by construction; its use is to point at an interaction d_{ij_r} between i and a unit j_r added on the way from S_1 to S_2 . A reversal between *incomparable* contexts does not contradict submodularity: on $V = \{i, a, b\}$ with $f(\emptyset) = f(a) = f(b) = f(ab) = 0$, $f(i) = 2$, $f(ai) = -1$, $f(bi) = 1$, $f(abi) = -2$, all six second differences are nonpositive, yet $\Delta_i f(\{a\}) < 0 < \Delta_i f(\{b\})$. That submodular example also contains the nested $d \rightarrow s$ reversal $\emptyset \subset \{a\}$, as Proposition 1 allows.

Proposition 2 (preference drift). *For a chain as in Lemma 1 and $i, j \notin S_m$, $(\Delta_i f - \Delta_j f)(S_m) - (\Delta_i f - \Delta_j f)(S_0) = \sum_{r=1}^m (d_{ik_r} - d_{jk_r})(S_{r-1})$, with k_r the added units. If the preference between i and j reverses materially, every fixed ranking errs in one of the two contexts, and some $d_{ik_r} - d_{jk_r} \geq 2\tau/m$.*

Proof: Apply Lemma 1 to i and j and subtract. ■

Remark 1. If $f(S) = h(|S|) + \sum_{k \in S} w_k$ then $d_{ik} = d_{jk}$ and the ranking by w is exact in every context, whether h is concave or convex. Heterogeneity of interactions across units is thus necessary for a fixed ranking to fail; the sign of the interactions is not what matters. Sub- and supermodularity refer to f on all of 2^V ; the stable portfolios do not form a lattice, so no greedy guarantee is at stake.

IV. DYNAMIC INTERVENTION SENSITIVITY

Let $w = (x, z, r)$ collect states, bus voltages and freed references r . Under *scheduled-power-and-voltage* (SPR) semantics the references of every unit are re-solved so that it keeps its scheduled active power and terminal voltage; the slack machine at bus 39 absorbs the loss change and fixes the angle reference, and the frequency is the nominal one, which pins the symmetry directions. The equilibrium solves $R(w; a) = 0$. For a simple critical eigenvalue λ with right and left eigenvectors u, v of A ,

$$\frac{d\lambda}{da} = \frac{v^H (\partial_a A + \partial_w A[\dot{w}]) u}{v^H u}, \quad \dot{w} = -R_w^{-1} R_a \quad (2)$$

(implicit function theorem). The first term alone is the *frozen* derivative D^{fro} ; the full expression is the *total* derivative D^{tot} . This is the feasible-sensitivity construction of [21], [22] applied to portfolio-specific operating points; no novelty is claimed for it.

Remark 2. If a is a controller gain of a unit whose scheduled output is fixed (the GFL gain g_i , a PLL scale, an AVR gain), the SPR states (x^*, z^*) do not move; only the freed references move, and they enter f and g affinely, so $\partial_w A[\dot{w}] = 0$ and $D^{\text{tot}} = D^{\text{fro}}$. Voltage setpoints enter only through references, so their frozen derivative is zero. Branch admittances and loads move (x^*, z^*) .

A branch intervention scales the complete two-port admittance of branch e by γ_e (tap unchanged). The score is $-d\alpha_{\perp}/d\gamma_e$ at $\gamma_e = 1$; the truth is the fully re-solved change $-\alpha_{\perp}(\gamma_e) - \alpha_{\perp}(1)$ at $\gamma_e \in \{1.10, 1.25, 1.50\}$. D^{tot} has no fitted parameters, so “held out” qualifies only the comparison baselines. The static baselines are branch flows $|P_e|, |S_e|$; impedance $|z_e|$; endpoint electrical distance on the subtransient bus impedance matrix; effective resistance and Fiedler edge score on the coupling Laplacian L_B ; weighted edge betweenness; endpoint dV/dQ ; and the change of the generalized SCR of the target portfolio at $\gamma_e = 1.5$. The static indices are computed on the base network and power flow and therefore do not change across conditions; we add, post hoc, a data-driven fixed list (the mean finite-effect rank over the conditions of the other clusters).

V. EXPERIMENTAL DESIGN

Policies and holdouts. Policies lie in $u \in [0, 1]$ ($g = u^2$), $k \in [0.5, 2.3]$, $t \in [0.5, 3.0]$, $h \in [0, 2]$. Fifteen *discovery* policies lie on axis-aligned lines through P4; an *old holdout* of 24 Latin-hypercube points was used in a first campaign; the *new holdout* N01–N24 is a maximin Latin hypercube (best of 2000 designs, seed 20260929) fixed before any computation and never screened for stability. Policies whose all-SG base is unstable (5 new, 6 old) are kept and excluded from tests that need a stable base.

Envelopes. Four envelopes perturb machine groups fleet-wide (EM-f) or per unit (EM-u), converter groups (EC), or both (EMC). Ten fresh Latin-hypercube draws per envelope are evaluated at P4; they are held out in parameters, not in policy.

Computations. All 512 portfolios of V_9 are solved with mode shapes at every policy (32 256 portfolio evaluations). Branch sensitivities are computed at 88 new-holdout conditions for all 46 branches; the implicit-function derivatives match small-step finite differences within 5% (or 10^{-4}) in 100% of 4048 cases. Topology is evaluated at 16 policies for 35 admissible outages and 46 doublings. Reruns of selected tasks were identical in every summary field.

Cross-model holdout. No validated grid-forming model exists in our code base. The one validated alternative dynamic converter is a WECC library GFL chain (PLL, REGC, REEC-B and REPC-A blocks [38] in ANDES [33]) whose parameters were frozen and validated against an EMT implementation in earlier work on a different IEEE 39-bus configuration. We attach it unchanged (constant- Q mode) to a transcription of our machine models that reproduces the internal $\alpha_{\perp}$ of the all-SG base within $2 \times 10^{-7} \text{ s}^{-1}$ at every tested policy (ALT-WECC); the composition itself is validated only at that base and by initialization residuals. Three integrators per converter that belong to disabled or unused control paths and are structurally decoupled in the reduced state matrix are removed before the stability verdict; these rules were fixed after one matrix case had been seen (disclosed in the deviation log).

Statistics and preregistration. Policies and draws are designed points, so fractions are coverages of the tested conditions and intervals are percentile bootstraps over policy (or envelope) clusters. Three secondary tests form a Holm family: T1 (sign test of $p^* < 0.90$, raw $p = 1.9e - 06$, Holm $5.7e-06$), T2 (paired advantage of D^{tot} over $|P_e|$ minus 0.20, sign-flip, raw and Holm $p = 0.061$ on 6 clusters) and T3 (mixing, below). All gates are effect-size thresholds frozen in the preregistration. Post-hoc analyses requested by internal reviewers are labelled as such.

VI. RESULTS ON THE IEEE 39-BUS SYSTEM

A. Nested reversals

Fig. 1 shows the strongest level-D nested reversal on the new holdout. At N24, replacing unit 33 in $S_1 = \{31\}$ moves the tracked 0.64 Hz mode right by 0.053 s^{-1} ; in $S_2 = \{31, 34, 38\}$ the same replacement moves the same family, now at 0.74 Hz (MAC 0.97 between contexts), left by 0.042 s^{-1} . Table II gives coverage by level. At the preregistered primary level C,

TABLE II

COVERAGE OF REVERSAL ACROSS THE TESTED BASE-STABLE POLICIES AT $\tau_{\text{mat}} = 0.01 \text{ s}^{-1}$. LEVEL C IS THE PREREGISTERED PRIMARY GATE (≥ 0.75 ON THE NEW HOLDOUT).

reversal type	discovery	old holdout	new holdout
any stable-context pair	15/15	18/18	19/19
A: nested, global $\alpha_{\perp}$	15/15	18/18	19/19
B: nested, tracked mode	14/15	16/18	18/19
C: B, EM-band critical modes	14/15	16/18	17/19
D: C, same mode in both contexts	14/15	16/18	17/19

New holdout by direction ($s \rightarrow d$ / $d \rightarrow s$ / both): level C 9/15/7, level D 6/14/3 of 19. Excluding pairs whose rightmost-eigenvalue gap is below τ_{mat} at any of the four portfolios (post hoc): level C 16/19, level D 16/19. Base-unstable policies (kept, excluded): discovery 0, old 6, new 5.

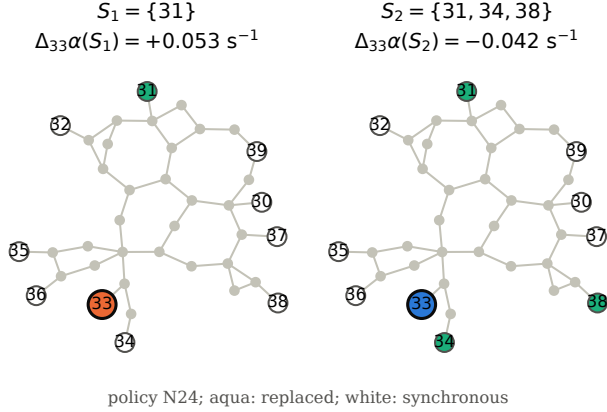


Fig. 1. The same replacement (unit 33) in two nested stable portfolios at new-holdout policy N24 moves the same tracked EM mode in opposite directions (orange: destabilizing; blue: stabilizing). Aqua: replaced units; white: synchronous.

17/19 base-stable new-holdout policies have a nested reversal (coverage 0.89; the gate ≥ 0.75 passes on the point estimate, and the bootstrap interval 0.74–1.00 reaches below it); the same policies have one at level D. Excluding pairs whose rightmost eigenvalue is within τ_{mat} of the next at any of the four portfolios leaves 16/19. Following the critical mode of S continuously while the added unit is replaced fractionally ends on the MAC-matched mode in 100% of 60 tested endpoint marginals. Fig. 2 shows the three strongest level-D pairs for distinct units.

The effects are small. Over the 3067 level-C pairs of base-stable new policies (87 distinct unit-policy combinations), reversal magnitudes $\min(|\Delta_i f(S_1)|, |\Delta_i f(S_2)|)$ have quartiles 0.011–0.017 s^{-1} , or 0.27–0.39% of damping ratio, and the largest is 0.042 s^{-1} . Coverage falls with the threshold (Fig. 3, right): at 0.02 s^{-1} level C holds in 15/19 and level D in 11/19 policies, at 0.03 s^{-1} in 10/19 and 5/19, and at 0.05 s^{-1} in none. Destabilizing $\rightarrow$ stabilizing reversals dominate: at level D, $s \rightarrow d$ occurs in 6 and $d \rightarrow s$ in 14 policies, and both directions on one tracked mode in 3. Converter-parameter draws alone (EC) remove every EM same-mode reversal from the four-unit lattice at P4 (below).

By Proposition 1, $s \rightarrow d$ reversals exclude submodularity and $d \rightarrow s$ reversals supermodularity; both directions occur at level A in 17 and at level C in 7 of 19 policies. The same conclusion

TABLE III

FIXED AND SCREENED NEXT-REPLACEMENT RULES (MEDIAN OVER BASE-STABLE POLICIES). ROWS 1–3: AGREEMENT-OPTIMAL FIXED ORDER; PREREGISTERED INSUFFICIENCY BAR $p^* \leq 0.90$ AND REGRET RATE ≥ 0.10 . ROWS 4–6 ARE POST HOC. $\dagger$ FIRST RANKED UNIT WHOSE NEXT PORTFOLIO IS STABLE, ON CONTEXTS WITH AT LEAST ONE STABLE OPTION.

rule	old holdout p^*	old holdout regret	new holdout p^*	new holdout regret
all transitions	0.75	0.37	0.75	0.38
EM $\rightarrow$ EM transitions	0.97	0.03	0.97	0.03
tracked EM mode (level C)	0.98	0.02	0.98	0.02
regret-optimal order, all	–	0.34	–	0.34
stability-screened order $\dagger$	–	0.07	–	0.06
max gSCR of $S \cup \{i\}$ (static)	–	0.77	–	0.76

follows directly from one-step squares whose four portfolios are stable with EM critical modes: second differences of both signs beyond 10^{-3} s^{-1} occur in 18/19 new and 16/18 old base-stable policies (Fig. 3, left). Fig. 4 maps how often each replacement destabilizes; no unit is uniformly stabilizing or destabilizing.

B. Fixed rankings and their anatomy

The fixed ranking of the nine units that maximizes pairwise agreement with a policy’s own resolved preferences is found exactly by dynamic programming. Table III and Fig. 5 report it. Over all transitions it orders 0.75 of resolved pairs on the new holdout and its next choice is at least τ_{mat} worse than the context-aware choice in 0.38 of stable contexts (19/19 policies have $p^* < 0.90$); the regret-optimal fixed order, found by a dynamic program over placed prefixes, reaches 0.34. The old-to-new shift is -0.002 in p^* . A ranking fitted at one policy errs in 0.40 of decisions at another, against 0.38 in sample: the failure is present in sample and transfer adds little (median Kendall between the policies’ optimal orders 0.72).

The errors have one source. In 2089 of the 2265 materially wrong decisions (92%) the ranked unit pushes the rightmost eigenvalue out of the EM band; in 22% of the contexts no candidate replacement keeps the portfolio stable. These out-of-band critical eigenvalues are real in 99% of cases, with rates of 20–3456 s^{-1} (5th–95th percentile), and appear when five to seven of the nine units are converted. Their participation concentrates on the converters’ PLL angle (0.39) and q-axis current (0.36). Replacing the added unit fractionally, $\alpha_{\perp}$ jumps within one 0.025 step of the replaced fraction from about zero to 3773–109804 s^{-1} , exactly where the smallest singular value of g_z reaches 0.002–0.007 (six sweeps, supplement): the signature of a singularity-induced bifurcation of the DAE [39]. These events mark a loss of the index-one property of the phasor model at high converter share; their physical course would require network dynamics or an EMT model, which were not used. Restricted to transitions that stay in the EM band, the oracle ranking orders 0.97 of pairs and errs in 0.03, failing the preregistered insufficiency bar; screening the ranking for stability (first ranked unit whose next portfolio is stable) lowers its regret from 0.27 to 0.065 on contexts with a stable option. A portfolio-conditioned static screen, choosing

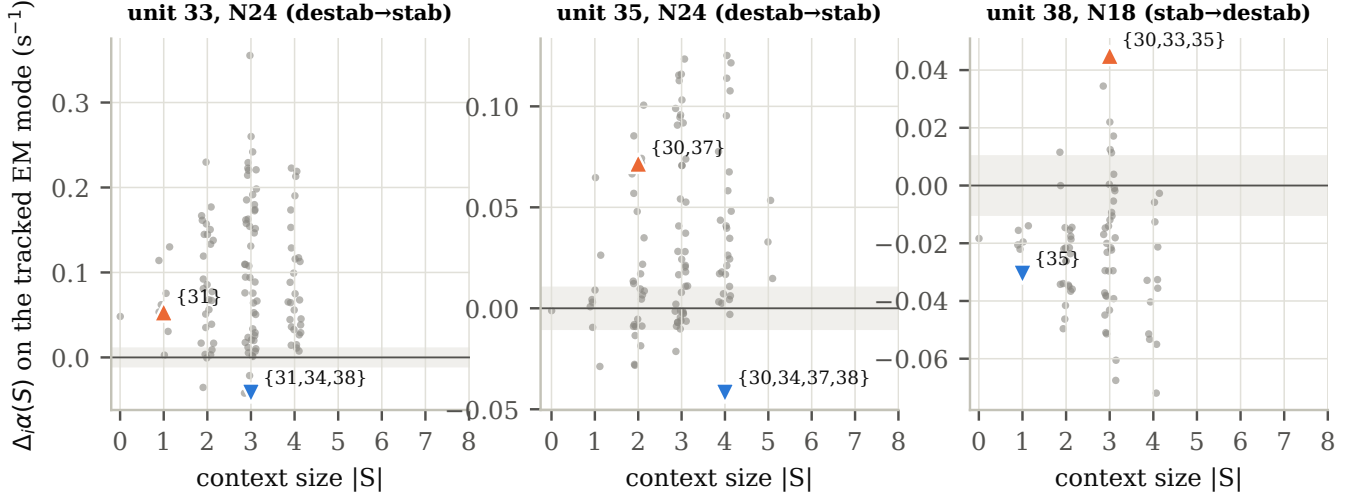


Fig. 2. Strongest level-D nested reversals on base-stable new-holdout policies (distinct units). Gray: the unit’s level-C contexts; triangles: the nested pair; band: $\pm\tau_{\text{mat}}$.

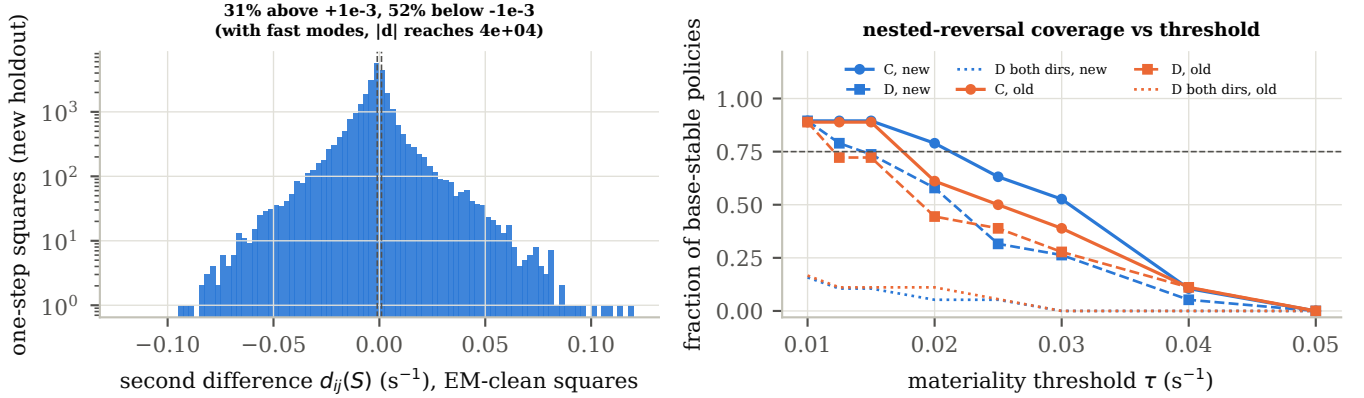


Fig. 3. Left: second differences over one-step squares whose four portfolios are stable with EM critical modes (new holdout). Right: nested-reversal coverage against the materiality threshold (level C, level D, and level D in both directions).

the unit that maximizes the generalized SCR of $S \cup \{i\}$, errs in 0.76 of decisions.

The preregistered evidence layers (Table IV) meet the frozen rule for the claim that a context-independent node ranking does not suffice (7 of 8 positive), but the layers are nested and the ranking layers do not survive the EM restriction; we read the result as: in this model a fixed list fails because it cannot anticipate which replacement crosses the singularity, and within stable EM dynamics it is a reasonable approximation.

C. Portfolio-conditioned reinforcement ranking

Table V and Fig. 6 report branch ranking for the core portfolio V_4 . On the 8 preregistration-eligible conditions (equilibrium residual $\leq 10^{-8}$, a threshold below the solver’s residual of order 10^{-7} in most conditions; six of the eight are draws at P4), the median Spearman correlation with the finite $\times 1.5$ outcome is 0.985 for D^{tot} , 0.913 for D^{fro} and 0.491 for $|P_e|$, with paired advantage 0.492 (interval 0.477–0.608) and top-5 precision 1.000; the preregistered gate passes,

TABLE IV
PREREGISTERED EVIDENCE LAYERS FOR “A CONTEXT-INDEPENDENT NODE RANKING DOES NOT SUFFICE” (NEW HOLDOUT). FROZEN RULE FOR A STRONG CLAIM: ≥ 5 of 8 INCLUDING L2 AND L5. L1–L4 ARE NESTED, L6 IS POSITIVE WHENEVER L5 IS, L5 FAILS IN THE EM STRATUM, AND L7 USES THE GLOBAL $\alpha_{\perp}$, SO THE LAYERS ARE NOT INDEPENDENT.

layer	value	pos.
L1 arbitrary-context reversal	1.00	yes
L2 nested-context reversal (level A)	1.00	yes
L3 tracked-mode nested reversal (level B)	0.95	yes
L4 EM-band tracked nested reversal (level C)	0.89	yes
L5 optimal fixed ranking failure (FULL)	$p^*=0.749$; regret=0.376	yes
L6 ranking transfer failure	offdiag regret=0.402; diag=0.376	yes
L7 envelope robustness (fresh draws)	EC: 1.0, EM-f: 0.9, EM-u: 0.8, EMC: 1.0; EM mode: EC: 0.0, EM-f: 0.5, EM-u: 0.6, EMC: 0.7	yes
L8 dynamic-converter holdout (ALT-WECC reversal)	0.0 (MODEL-SPECIFIC)	no

while the secondary test T2 does not reach $p < 0.05$ with six

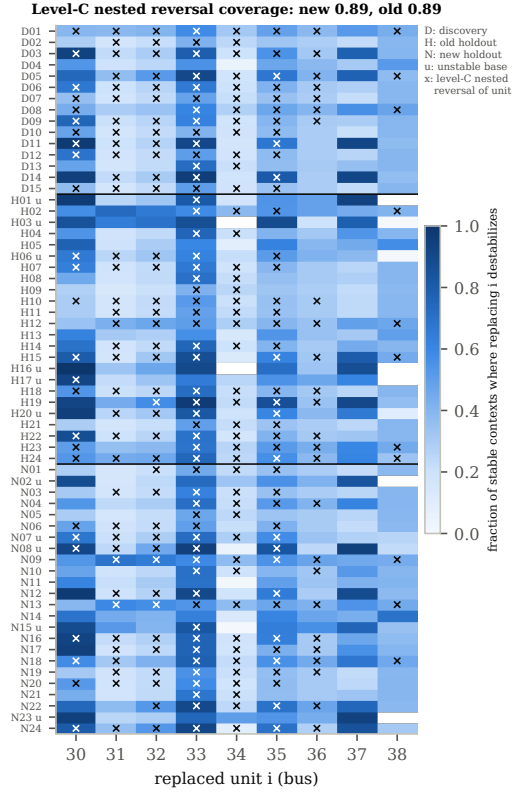


Fig. 4. Fraction of stable contexts in which replacing unit i destabilizes, by policy (D: discovery; H: old holdout; N: new holdout; u: unstable base). Crosses: level-C nested reversal of unit i (white on dark cells for contrast).

TABLE V

BRANCH-REINFORCEMENT RANKING FOR V_4 ON THE 24 NEW-HOLDOUT POLICIES AND THE 40 FRESH DRAWS AT P4 (TRUTH: FINITE RE-EQUILIBRATED $\times 1.5$). MEDIAN OVER CONDITIONS; “ELIG.”: THE 8 PREREGISTRATION-ELIGIBLE CONDITIONS ($R_0 \leq 10^{-8}$); OTHER COLUMNS: ALL 64 SOLVED CONDITIONS. THE PORT FORM RE-DERIVES THE TOTAL DERIVATIVE FROM RE-SOLVED JACOBIANS (ITS “ELIG.” ENTRY USES 2 CONDITIONS). † POST HOC: MEAN FINITE-EFFECT RANK OVER THE CONDITIONS OF THE OTHER CLUSTERS.

predictor	ρ elig.	ρ	τ	concord.	top-5	NDCG@5
$ P_e $ (selected on discovery)	0.49	0.50	0.39	0.71	0.20	0.61
$ S_e $	0.47	0.48	0.37	0.69	0.20	0.61
$ z_e $	0.25	0.24	0.20	0.61	0.20	0.27
electrical distance	0.21	0.19	0.15	0.58	0.40	0.38
effective resistance	0.30	0.30	0.22	0.61	0.60	0.58
Fiedler edge score	-0.05	-0.05	-0.03	0.48	0.20	0.36
edge betweenness	-0.28	-0.27	-0.18	0.40	0.00	0.17
endpoint dV/dQ	-0.17	-0.16	-0.10	0.45	0.00	0.16
$\Delta gSCR(V_4)$	0.16	0.17	0.12	0.57	0.20	0.30
base-portfolio eig. sens.	-0.05	0.03	0.01	0.50	0.60	0.91
frozen (portfolio)	0.91	0.90	0.77	0.90	0.60	0.97
total (portfolio)	0.98	0.99	0.93	0.98	1.00	0.99
port form (verification)	0.99	1.00	0.96	1.00	1.00	1.00
fixed list from other clusters †	0.95	0.94	—	—	0.80	—

clusters. Over all 64 solved conditions (residual $\leq 2.1 \times 10^{-7}$) the medians are 0.987, 0.895 and 0.497.

The informative comparisons are with D^{fro} and with a fixed list. Across the 24 new policies, D^{tot} has median Spearman 0.996, D^{fro} 0.712 and the learned fixed list 0.818; across the 40 draws at P4 the list reaches 0.959 against 0.986. D^{tot}

exceeds the list in every condition, but the median paired margin is 0.03 on the draws and 0.17 on the new policies, not the 0.5 gap to static indices, which do not respond to conditions at all. The branch ranking thus changes mostly across controller policies. D^{tot} also ranks large actions: at 6 new-holdout policies, Spearman with the finite effect of doubling a branch is 0.992 and of removing it 0.885 (top-5 0.80). The first-order prediction overstates the largest reinforcements, whose effect saturates at $\gamma_e = 1.5$ (Fig. 6, right). The port form in Table V re-derives D^{tot} from re-solved Jacobians and is a verification, not an independent method.

D. When re-equilibration matters

Splitting (2) into frozen and re-equilibration parts (Fig. 7): for g_i , PLL scales and AVR gains the two derivatives agree to $2.2\text{e-}07$ relative, as the remark predicts. For loads the re-equilibration term exceeds the frozen term in 61% of cases. For branches it exceeds the frozen term for about half of the branches and changes the ranking materially (sign disagreement $\geq 10\%$ or Kendall ≤ 0.6) in 28% of the V_4 conditions; every condition has a branch whose frozen and total signs differ.

E. Robustness to parameter draws

Under the fresh envelope draws at P4 the coverage of a stable-context reversal on the four-unit lattice is 1.00 (EC), 0.90 (EM-f), 0.80 (EM-u) and 1.00 (EMC); for EM same-mode reversals it is 0.00, 0.50, 0.60 and 0.70. The EM reversals are therefore sensitive to the converter parameters. Branch ranking on the draws gives median ρ 0.986 for D^{tot} and 0.503 for $|P_e|$.

F. Cross-model holdout

Table VI summarizes ALT-WECC on the seven preregistered policies with a stable ALT base. Its converted portfolios are less stable than with the custom GFL; V_4 is unstable at all policies. The preregistered test on the global $\alpha_{\perp}$ finds no stable-context reversal, but it is uninformative: in most stable ALT portfolios $\alpha_{\perp}$ is set at -0.100 s^{-1} by the anti-windup tracking state of the plant-level active-power integrator, a state decoupled from the rest of the model that our structural rule kept. On the tracked EM mode, stabilizing replacements occur at 3 of 7 policies and a nested reversal at 1 of 7 (N04), the same policy at which the custom GFL has a level-C reversal on this lattice. Switching on the library’s voltage control (all gains unchanged, post hoc) gives no stabilizing replacement and no reversal, so the data do not support the reading that converter voltage control is what enables reversals. The 12-branch finite ranking is model-specific (median Kendall 0.06), as a previous cross-tool check had indicated, while within ALT-WECC a finite-difference total sensitivity ranks the model’s own outcomes at 0.99 against 0.79 for $|P_e|$, a margin at the preregistered bar. Top-1 agreement of the equal-budget corridors (Section VI-H) is partial (0.71) and the topology directions agree (1.00 on 29 pairs).

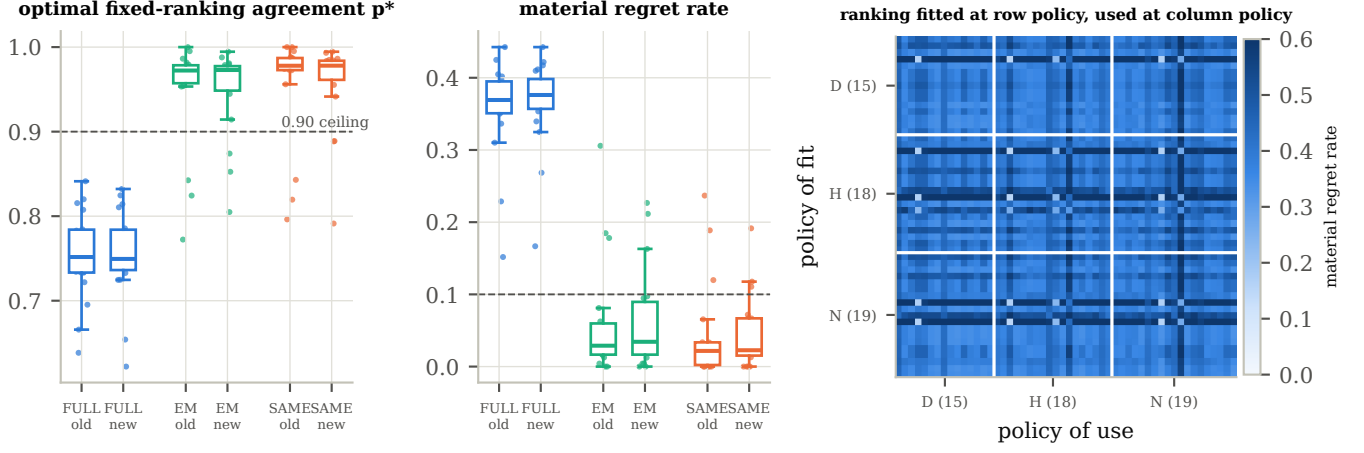


Fig. 5. Agreement-optimal fixed ranking: agreement p^* and material regret rate by stratum (old and new holdout), and regret of rankings transferred between base-stable policies (rows: policy of fit; columns: policy of use, ordered discovery, old, new).

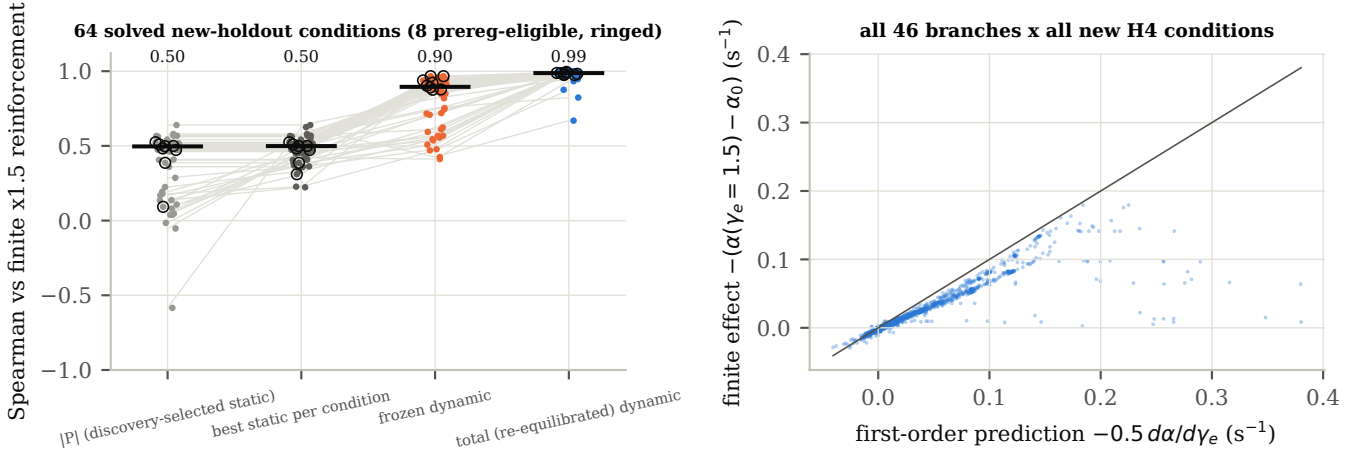


Fig. 6. Left: Spearman correlation with the finite reinforcement outcome per new-holdout condition (lines join the same condition; ringed: the preregistration-eligible subset). Right: first-order prediction vs finite effect, all branches and conditions.

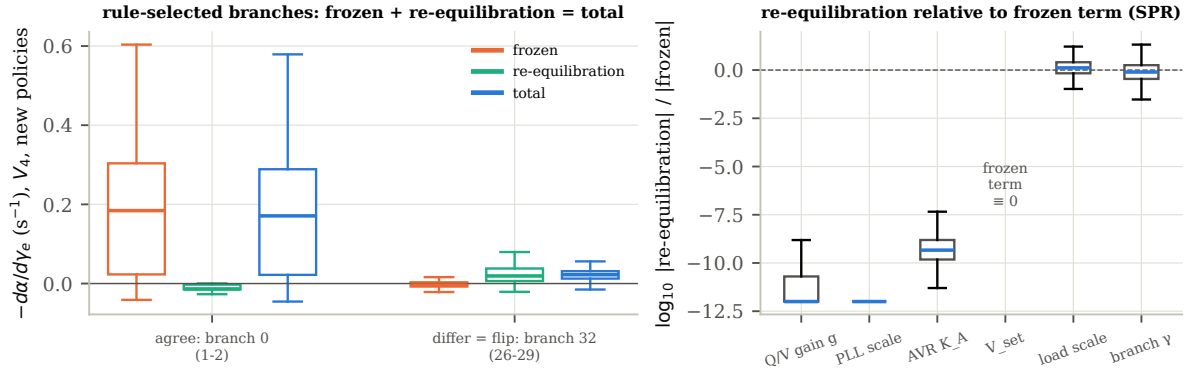


Fig. 7. Left: frozen, re-equilibration and total branch sensitivities for the branches selected by the preregistered rule. Right: ratio of the re-equilibration term to the frozen term by coordinate class (the frozen term of voltage setpoints is identically zero).

G. Topology and electromechanical modes

Topology effects were re-evaluated by tracking the pre-action EM-band rightmost mode of each portfolio through

every admissible action; events set by a non-EM mode (50 of 1296 V_4 policy-action cases) are excluded. At P4, doubling any one of lines 1–2, 1–39, 8–9 or the 23–36 transformer

TABLE VI
CROSS-MODEL HOLDOUT: FROZEN WECC LIBRARY GFL CHAIN (ALT-WECC) VS THE CUSTOM GFL ON THE 7 OF 8 PREREGISTERED POLICIES WITH A STABLE ALT BASE. IN ALT THE GLOBAL $\alpha_{\perp}$ OF MOST STABLE PORTFOLIOS IS PINNED AT -0.100 s^{-1} BY A DECOUPLED INTEGRATOR, WHICH MAKES TEST A UNINFORMATIVE.

test	value	verdict
A stable-context reversal, global $\alpha_{\perp}$	0.00 of 7 (custom 0.71)	model-specific
A' same, tracked EM mode (post hoc)	stabilizing at 3/7, nested reversal at 1/7	partial
A'' voltage control on (post hoc)	0/7 global, 0/7 tracked	none
C 12-line finite ranking, Kendall	median 0.06	model-specific
C2 ALT total vs $ P $ on ALT outcomes	0.99 vs 0.79 (3 of 7 below bar)	transfers
D corridor top-1 agreement	0.71	partial
E topology sign agreement	1.00 (29 pairs)	transfers

makes V_4 stable; across the 16 policies, 45 (policy, action) pairs create an EM-failing hyperedge of at most three units, all by outages. No static topology score meets the prediction rule ($|\rho| \geq 0.6$ in $\geq 75\%$ of policies), and within doublings their correlations are near zero (Fiedler value 0.00; supplement). This is expected N-1 small-signal behavior and we report it as a supporting observation.

H. Equal-budget corridors

A first campaign ranked groups of branches (corridors) by the effect of scaling every member by 1.5, which favors large groups. The preregistered test gives every frozen corridor the same budget $\sum_{e \in C} |\Delta\gamma_e| = 0.5$ and compares its stabilizing effect on V_4 with 500 arbitrary branch groups of the same size (family A; connected groups, family B, as a secondary null). A corridor is called robust if it lies in the top 5% of family A in at least 75% of the 64 new-holdout conditions. Table VII reports the test. The rule is met only by the group of the other eight transformers (the five generator step-up transformers outside V_4 and three network transformers), which is in the top 5% in 0.83 of the 64 conditions (16/24 new policies and 37/40 draws at P4; old holdout 0.78; connected null 0.84). The pass rests on the parameter draws at one policy: across controller policies alone the group would miss the 75% bar. The group of all 12 transformers reaches 0.69, and no spectral cutset exceeds 0.03.

I. Modal mixing is not confirmed

A correlation between controller-weighted modal mixing and the number of reversing units, reported by a first campaign, fails the preregistered test on the new holdout ($\rho = 0.55$, $p = 0.016$, Holm 0.033; supplement).

VII. REPLICATION ON THE IEEE 68-BUS SYSTEM

A. Model and design

The replication was preregistered before any converter portfolio was evaluated. The network is the 68-bus, 16-machine NETS-NYPS benchmark [40], [41]: 83 branches,

TABLE VII
FROZEN CORRIDORS UNDER AN EQUAL BUDGET ($\sum_{e \in C} |\Delta\gamma_e| = 0.5$) ON THE NEW HOLDOUT (24 POLICIES, 40 DRAWS): FRACTION OF CONDITIONS STABILIZED BY $\geq \tau_{\text{mat}}$, FRACTION IN THE TOP 3, MEDIAN EFFECT, AND FRACTION OF CONDITIONS IN THE TOP 5% OF THE SIZE-MATCHED ARBITRARY (A) AND CONNECTED (B) NULLS. GATE: A ≥ 0.75 .

corridor	$ C $	stab.	top-3	effect	null A	null B
other 8 transformers	8	0.67	0.99	0.035	0.83	0.84
all 12 transformers	12	0.65	0.85	0.028	0.69	0.77
step-up transformers of V_4	4	0.51	0.02	0.012	0.00	0.00
cutset 1-2, 3-4, 14-15	3	0.57	0.22	0.023	0.00	0.00
cutset 17-18, 17-27	2	0.00	0.01	-0.000	0.00	0.00
cutset 3-4, 9-39	2	0.53	0.10	0.018	0.00	0.00
line 14-15	1	0.00	0.01	-0.002	0.00	0.00
line 16-19	1	0.66	0.80	0.028	0.03	0.03

sub-transient machines with damper windings, DC4B, ST1A or manual excitation, speed-input stabilizers and constant-impedance loads. The published benchmark has no governors and $D = 0$. We add the governor and rotor-damping data documented in the Power System Toolbox case of the same system [42]: a steam-reheat governor (PST model 1, $1/R = 25$ on the machine rating) on every surviving machine, with equations from [43], and damping on the four area equivalents. The confirmatory variant (REAL) includes both. Two ablations remove the governors (NOGOV) or both (the published model, SP33). With governors the rotation symmetry leaves one structural zero, and the rightmost transverse eigenvalue of the all-SG base is the stabilizer washout pole at -0.067 s^{-1} ; the rightmost EM mode is the 0.555 Hz inter-area mode at -0.37 s^{-1} .

The candidates are the six physical plants with the largest scheduled output, $V_{68} = \{3, 4, 6, 9, 11, 12\}$ (64 portfolios), chosen by a stability-blind rule. Model A is the GFL of Section II, unchanged. Model B is the frozen WECC library chain of Section VI-F, attached in ANDES to a new transcription of the 68-bus machines, governors and damping. The transcription reproduces the internal $\alpha_{\perp}$ of the all-SG base within 10^{-10} s^{-1} at all 16 excitation scales. Sixteen maximin Latin-hypercube policies (g, k) were used, four for discovery and twelve as holdout; Model B uses their k . The reversal gate was 50% of base-stable policies with a level-D nested reversal at unchanged τ_{mat} . The ranking gate was the one of Section VI-C per model: median $\rho \geq 0.70$, paired advantage ≥ 0.20 over the discovery-selected static index, and top-5 precision ≥ 0.60 . The decision between pivoting and stopping followed a rule fixed in advance.

B. Results

Table VIII summarizes the replication. Every one of the $2 \times 16 \times 64$ portfolios is stable, and in all of them $\alpha_{\perp}$ is a slow real pole, never an EM mode. The global marginals are therefore immaterial, and levels A-D are empty. On the tracked EM mode (level T) the marginals of both converters are almost one-sided. They range from -0.0069 to 0.053 s^{-1} (A) and from -0.0008 to 0.057 s^{-1} (B). 437 and 1429 of them are materially destabilizing, and none is materially stabilizing. A reversal needs a stabilizing side, so none exists. Interaction

TABLE VIII

REPLICATION ON THE IEEE 68-BUS SYSTEM WITH PST GOVERNORS AND DAMPING (16 POLICIES PER CONVERTER MODEL; BRANCH RANKING ON 12 HOLDOUT CONDITIONS AT $\gamma_e = 1.5$)

	A: custom GFL	B: WECC GFL
level-D nested reversal	0/16	0/16
level-T (EM-tracked) reversal	0/16	0/16
EM-tracked marginals (s^{-1})	$[-0.0069, 0.053]$	$[-0.0008, 0.057]$
materially stabilizing marginals	0	0
fixed unit ranking, regret	0	0
median $\rho(D^{\text{tot}}, \text{finite})$	0.996	0.981
selected static index, ρ	0.20 (el. dist.)	0.43 (ΔgSCR)
median paired advantage	0.80	0.55
median top-5 precision	1.0	0.8
largest effect on $\alpha_{\perp}$ (s^{-1})	1.4×10^{-3}	3.4×10^{-5}

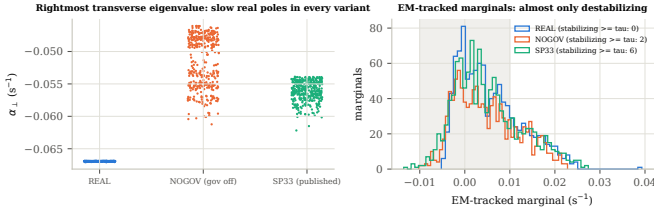


Fig. 8. IEEE-68, Model A: rightmost transverse eigenvalue of all portfolios of six policies (left) and EM-tracked replacement marginals (right) with governors and damping (REAL), without governors (NOGOV) and in the published model without either (SP33). Band: $\pm\tau_{\text{mat}}$.

terms of both signs are present, but they change only the size of a destabilizing marginal. The same holds without governors (0/6, 2 stabilizing marginals) and in the published model without governors or damping (0/6, 6 stabilizing marginals, the smallest $-0.0136 s^{-1}$), and in 40 and 20 deterministic parameter draws of ± 5 and $\pm 10\%$ (reversals in 0/40 and 0/20). An oracle fixed unit ranking has zero regret on the tracked mode, so the fixed-ranking failure of IEEE-39 does not replicate either (Fig. 8).

For branch reinforcement the implicit-function total derivative matches small central differences with median relative error 1.3×10^{-5} (Spearman 0.99995, 80 branch-policy pairs). On the twelve holdout conditions it ranks the finite $\times 1.5$ effects with median Spearman 0.996 (A) and 0.981 (B). The static comparators selected on the discovery policies reach 0.20 (electrical distance) and 0.43 (ΔgSCR). The paired advantages are 0.80 and 0.55, and the top-5 precisions 1.0 and 0.8. On ten parameter draws at one policy the Model A median stays at 0.996. The frozen derivative reaches 0.55 and the base-case total derivative 0.53 (Fig. 9). For Model B no analytical engine exists, and its total derivative is a re-equilibrated central difference. The advantage test T2 gives $p = 0.0078$ (Holm 0.016) for A and $p = 0.090$ for B. The latter is not significant, because the median sign-flip statistic has little power when the twelve advantages are nearly equal.

These rankings concern a pole that the reinforcements barely move. The median over conditions of the largest finite effect is 4.2×10^{-5} (A) and $3.3 \times 10^{-5} s^{-1}$ (B), far below τ_{mat} . Projecting the same perturbed matrices on the rightmost EM mode (post hoc) gives median Spearman 0.998 and top-

5 precision 1.0 for Model A. The largest effects there are material, with a median per condition of $0.020 s^{-1}$ and a maximum of $0.047 s^{-1}$. The two models rank the same branches differently: between A and B the finite effects correlate with median Spearman 0.50 (top-5 overlap 0.4). A nonlinear phasor simulation of the preregistered disturbance (a 0.5 pu reactor at bus 3 for 10 s) could not be run as preregistered, because no reversal exists. Run post hoc on 2 same-sign nested pairs, it reproduces the linear decay changes within $3.6 \times 10^{-6} s^{-1}$, with the same sign in 4 of 4 cases.

VIII. DISCUSSION

What a planner needs. For reinforcement, the re-equilibrated sensitivity of the planned portfolio ranked finite reinforcements on both benchmarks and under both converter models, and on IEEE-39 also doublings and outages. The frozen derivative degrades wherever branch actions move the operating point, and neither the static indices nor the exact line list carry over between converter models. The ranking should therefore be recomputed for the portfolio, controller settings and converter model in which it is applied. Whether the ranked quantity matters is a separate question: on IEEE-68 the rightmost transverse eigenvalue is a control pole that reinforcements barely move, and a planner should rank the mode of interest, here the rightmost EM mode. For the choice of the next unit, IEEE-39 and IEEE-68 disagree. On IEEE-39 a fixed weak-bus list is unreliable because it cannot anticipate which replacement crosses the high-share singularity; on IEEE-68 no singularity is reached and a fixed list is exact on the tracked mode. We do not claim a computational saving: our implementation evaluates the total derivative with finite-difference Jacobians at 1.46 s per branch, against 0.61 s for a finite re-solve and eigenanalysis; a saving would require analytic second derivatives and large systems, which we did not test.

Relation to strength metrics and placement. Static indices are computed from the base network and do not respond to the policy or the draw; they carry some information (effective resistance places three of the five strongest branches in its top five) but do not order reinforcements within a portfolio. Placement methods built on grid-strength or Kron-reduced-Laplacian surrogates treat converters as homogeneous and, for grid-forming conversions, can show that each conversion helps [44]–[46]; the reversals here arise with heterogeneous machines and a converter with its own Q/V control, and they were not reproduced by the library converter. Offline priority lists of switching actions [47] are fixed rankings of the kind examined in Section VI-B.

Literature scope. The gap statements rest on a focused search of 2020–2026 journal and conference records, extended to the earlier displacement literature, with arXiv only where no peer-reviewed version exists and without full-text database access; they state the absence of such results within that search.

Limitations. All results are phasor-domain small-signal results on two benchmark networks with two grid-following converter models; no grid-forming model was tested. The IEEE-39 machines have $D = 0$ and no governors and its loads are

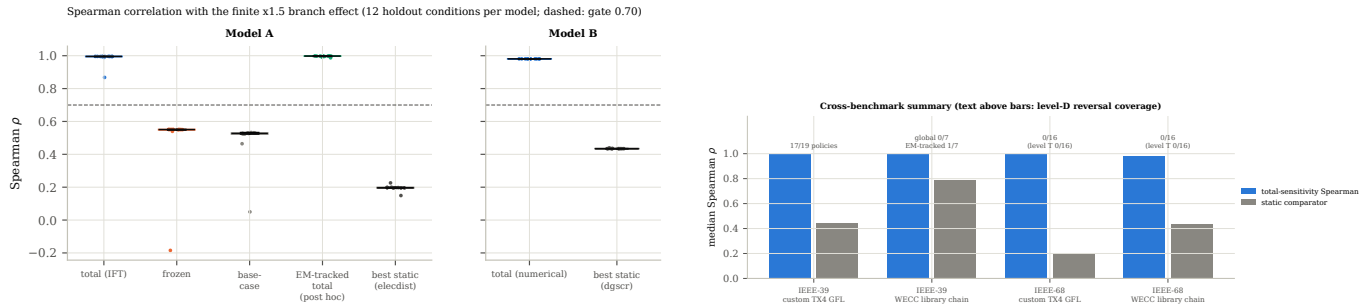


Fig. 9. Left: IEEE-68, Spearman correlation of each predictor with the finite $\times 1.5$ branch effect on the twelve holdout conditions (EM-tracked total: post hoc). Right: both benchmarks and both converter models, median Spearman of the total sensitivity and of the static comparator; text: level-D reversal coverage.

constant power; the IEEE-68 case adds documented governors, damping and impedance loads, and there the reversals are absent with and without them, so the absence is a property of that network and not of the added dynamics. On IEEE-68 the preregistered ranking target is an immaterial control pole; the material EM ranking is post hoc and needs its own preregistered confirmation. The fast events behind the IEEE-39 fixed-ranking failure are singularity crossings of an idealized phasor converter without current limits. The primary GOLD-B set on IEEE-39 has eight conditions because the preregistered residual threshold was stricter than the solver tolerance. The deviation logs record execution fixes, post-hoc analyses, a defect in the cross-model tests found by internal review, clock times corrected against the commit history, and a corrected interpretation note of the IEEE-68 ablation.

IX. CONCLUSION

The re-equilibrated eigenvalue sensitivity of the planned converter portfolio ranked finite branch reinforcements better than static strength indices on the IEEE 39- and 68-bus benchmarks and under two converter models, with median Spearman correlations of 0.98 to 1.00 on held-out policies. The exact line list and the best static comparator changed with the converter model, and on IEEE-68 the preregistered target was a control pole that reinforcements barely move. The contextual findings of IEEE-39 did not transfer. Nested reversals on the same electromechanical mode, about one hundredth of s^{-1} there, are absent on IEEE-68 with both converters, with and without governors and damping, and a fixed unit ranking is exact there. Reinforcement rankings should be recomputed for the portfolio, controller settings and converter model in which they are applied, and for the mode that matters.

DATA AND CODE AVAILABILITY

Preregistration, deviation log, code, result tables and figure sources are in the project repository, branches `research/contextual-dynamic-weakness-hardening` (IEEE-39) and `research/cdw-ieee68-replication` (IEEE-68); software: Python 3.13, NumPy 2.5, SciPy 1.18, ANDES 2.0.

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